

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Product Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built production automation that connected Node.js services, browser workflows, API lookups, operator validation, Docker, Cloud Run, and GitHub Actions.
- Handled unreliable external workflows with retries, session recovery, state validation, and visible checkpoints, reducing manual filing time by **85%**.
- Worked close to product constraints where generated drafts had to remain inspectable before downstream automation executed.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built scientific Python tooling for MRI segmentation across **95 anatomical regions**, handling preprocessing, orientation correction, model inference, masks, and 3D visualization.
- Reduced runtime by **87%** using asynchronous and parallel execution over large medical-image volumes while preserving inspectable outputs.
- Debugged data/model-output issues inside a mature medical-imaging app across VTK, wxPython, NumPy, PyTorch, and ONNX-adjacent paths.

Open Source

[Contributions](#) 

- Contributed to research-adjacent open source including **Invesalious** medical-imaging workflows, **IOOS** scientific-data tooling, and **Tiled** visualization code, plus compatibility and security-validation fixes.

Projects

Seaweed: Contest Platform | *Go, TypeScript, React, PostgreSQL, Redis, Judge0, Docker, Kubernetes*

[GitHub](#) 

- Built product and platform paths for registration, submissions, sandboxed code execution, admin review, live rankings, and durable result records across contest traffic.
- Designed async verdict polling, PostgreSQL result schemas, Redis-backed runtime state, Docker/Kubernetes deployment, and **sub-5s P95 verdict latency**.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

[GitHub](#) 

- Built browser-automation-backed product workflows with REST APIs, RabbitMQ workers, Redis autosave, PostgreSQL persistence, durable job status, and recoverable publishing tasks.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so long-running API/browser workflows stayed observable and fault-tolerant.

Hyoka | *Python, FastAPI, PostgreSQL, Next.js, TypeScript, OpenTelemetry, Docker*

- Built backend and dashboard paths for operating agent behavior, including trace ingestion, suites, candidates, validation runs, promotion state, audit history, and gate decisions.
- Designed project-scoped API keys, worker leases, signed manifests, JSON schema checks, replay artifacts, and containerized Postgres-backed services.

Technical Skills

Languages: Python, TypeScript, Go, SQL, C++

Frameworks: PyTorch, FastAPI, React, Next.js, VTK, wxPython, Echo

Tools: NumPy, ONNX, PostgreSQL, Redis, Docker, GCP, GitHub Actions, OpenTelemetry, Prometheus, Judge0